nature human behaviour

ARTICLES

https://doi.org/10.1038/s41562-020-0853-6

Establishing the structure and replicability of personality profiles using the HEXACO-PI-R

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Previous attempts to identify personality profiles in the five-factor and HEXACO models of personality have produced inconsistent results. Here, using data from four independent samples, each with approximately 90,000 international respondents to the 100-item HEXACO Personality Inventory-Revised (HEXACO-PI-R), we demonstrated that a five-profile solution fit the data well. Exploratory analyses suggested that this solution was also consistent across gender and age groups. The five-profile structure replicated well with larger subsamples, but could not be reproduced consistently with samples of fewer than 500 individuals. However, even with small samples, the five-profile structure could be applied using the parameters obtained with the larger samples. We used HEXACO theory along with agency–communion and attachment theories to offer preliminary explanations and labels for the five profiles. We discuss how these theories, combined with parameter estimates provided by our research, can be used to generate and test hypotheses to validate the five-profile structure and evaluate its utility for personality research and other applications.

For several decades, one of the primary goals of personality psychology was to identify the general structure of human personality[1,2]. Using the lexical hypothesis of personality as their guide[3], researchers conducted psycholexical studies around the globe using variable-centred techniques such as factor analysis. Results of these early studies in English produced five general factors (that is, the Big Five and the five-factor model of personality (FFM)): openness to experience, conscientiousness, extraversion, agreeableness, and emotional stability[4–7]. Later studies with larger sets of trait descriptors and conducted in languages other than English resulted in a somewhat different factor structure including the addition of a sixth factor of personality (honesty–humility), leading to the establishment of the six-factor HEXACO model of personality[1].

The primary difference between the HEXACO model of personality and the FFM is the inclusion of the sixth factor honesty–humility, which explains additional variance in outcomes beyond the FFM[1,8–10]. In addition, emotional stability and agreeableness from the FFM are reorganized and, along with honesty–humility, are intended to reflect different elements of altruism. On the basis of the realignment (rotation) of the axes in the HEXACO trait space, the label emotionality replaces emotional stability in the FFM, and is intended to reflect a tendency to prevent physical and emotional harm to oneself and others. Similarly, agreeableness is now opposite to anger and reflects patience and how individuals may respond to situations when they are treated poorly or unfairly.

Much like variable-centred approaches were used in previous research to establish the number and structure of personality traits, an increasing number of researchers have begun to employ a person-centred approach to identify the number and configurations of personality profiles. Early studies involving the FFM generally found evidence for three profiles, commonly referred to as the ARC types (reflecting the names of the lead authors of the original papers)[11–13]. The three profiles—resilient, undercontrolled and overcontrolled—were interpreted and labelled on the basis of Block's self-regulation theory of personality[14,15]. Although some subsequent studies involving the FFM and HEXACO yielded four or five profiles[11,16–18], many continued to apply the same theoretical rationale and used variants of the ARC labelling scheme. More recently, it has been noted that the same labels are often applied to what appear to be quite different trait patterns[16,17,19]. This raises concern about both the labelling scheme and the appropriateness of the theoretical interpretation and meaningfulness of the profile approach.

If the profiles identified in person-centred studies are indeed stable and meaningful, they should replicate across studies[16,19,20]. Herzberg and Roth[19] noted that most person-centred studies have been conducted with relatively small ($n < 500$) and frequently quite unique samples. Although this might help to explain the inconsistency across studies, it remains to be determined whether a stable and reproducible set of profiles actually exists. Gerlach et al.[17] recently attempted to address this issue by examining the reproducibility of four profiles derived from the FFM across four large samples. They applied Gaussian mixture modelling to the data from one sample and then attempted to recover the same four FFM profiles from the remaining samples. They identified the same four profiles in one sample, but only three of the four profiles emerged in the other two, thus presenting some evidence for their reproducibility.

To date, only a few studies have directly addressed the issue of profile replicability across samples[16] and time[18] for the HEXACO model of personality. Daljeet et al.[16] obtained the same five-profile structure across two samples of students and Isler et al.[18] found the same four profiles over one- and two-year periods in a sample of voters from New Zealand. Although the ability to replicate a profile structure is an important criterion in the evaluation of validity[20,21], both sets of researchers acknowledged that, owing to the uniqueness of their samples, their findings might not generalize to other samples or reflect the profile structure of the HEXACO dimensions in the general population.

The primary purpose of the present study was to determine whether a stable and generalizable profile structure of the HEXACO dimensions could be established using latent profile analysis (LPA) applied to very large and diverse samples of respondents.

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We focused on the HEXACO framework because of the evidence provided in variable-centred research for the value added by inclusion of the honesty–humility dimension[1,9,22,23]. Because of this and other differences in the trait dimensions, the personality profiles reflected in the HEXACO personality space could differ from those obtained for the FFM.

The present research was conducted using data from more than 370,000 individuals from a wide range of ethnicities, nationalities, ages and gender who completed the English version of the 100-item HEXACO-PI-R. We posit that this sample should be sufficiently large to provide a reasonable approximation of the English-speaking population of HEXACO test takers. Accordingly, the results of the present study could reflect the full set of profiles that exist in the HEXACO space for this population, rather than a subset that might emerge using smaller and more restricted samples, such as the four profiles found by Isler et al.[18] and five profiles found by Daljeet et al.[16]. Although we do not believe that there is currently a robust theoretical basis from which to predict the precise number or configuration of HEXACO profiles present in the population, extrapolating from previous empirical results suggests that at least five profiles will emerge, and that the same profiles will emerge across various samplings of the present data. Therefore, we offered the following hypotheses:

- Hypothesis 1a: the number of profiles we extract from our 'population-like' dataset will be five or greater.
- Hypothesis 1b: the same number of profiles with the same configurations of dimensions will emerge across random subsamples drawn from a population-like dataset.
- Hypothesis 1c: the same number of profiles with the same configuration of dimensions will emerge across independent samples of comparable size to the original dataset.

We also tested a proposition derived from Herzberg and Roth's argument that differences in profile structure observed across studies may be due to the use of small samples that are not truly representative of the population[19] ($n < 500$). Accordingly, we hypothesize that:

- Hypothesis 2: the profile structure observed for the 'population' will replicate only in sufficiently large random samples.

Although it is difficult to know in advance at what point the ability to reproduce the population structure will break down, on the basis of arguments by Herzberg and Roth[19], we expect that it might be at a sample size of approximately 500.

Results

Identifying high-potential profile solutions (stage 1). In the first stage, we conducted LPA on the initial sample ($N = 90{,}289$) to generate solutions ranging from 3 to 10 profiles and identified which of these solutions provided a good fit to the data by inspecting elbow plots of various model-based statistical indicators (see Methods for details on these indicators). The point at which the slopes flatten can be taken as an indicator of the optimal profile solution. Table 1 presents the resulting values for these statistical indicators and the elbow plots are presented in Fig. 1.

Inspection of the elbow plots for the statistical indicators revealed a continuous improvement in the indicators from the three-profile solution to the ten-profile solution, without a distinct elbow point. We also examined the mean trait levels for each profile across the various solutions to search for patterns in the shape of the profiles extracted. In some cases, extracting an additional profile (for example, the transition from five profiles to six profiles) resulted in the duplication of an existing profile (that is, similar shape) but with a slight difference in elevation (mean level across traits). In other cases, extracting additional profiles resulted in the emergence of new, qualitatively distinct profiles. Qualitative differences across solutions (that is, differences in profile shape) is a criterion that has been suggested for use in determining profile meaningfulness[24,25]. Therefore, after close examination of the profile structures, we determined that the five-, seven- and nine-profile solutions all yielded new qualitatively distinct and potentially meaningful profiles.

Next, we examined the size of the profiles and entropy for each solution. LPA is probabilistic in nature; results therefore provide an estimate of the likelihood that each participant is best represented by a given profile within the solution (that is, posterior probabilities). In this context, profile size refers to the estimated proportion of the sample in each profile when individuals are classified into a single profile on the basis of their highest posterior probability. Profiles that contain less than 5% of the sample have been proposed to be too small to be robust[26]. Entropy is an index of classification accuracy, ranging from 0 to 1, such that higher values indicate that there is less ambiguity in the solution's determination of individuals' posterior probabilities. In both the five-profile and seven-profile solutions, the smallest profiles contained at least 5% of the population, and the entropies for these solutions were 0.559 and 0.558, respectively. By contrast, the nine-profile solution included a profile representing only 3.2% of the sample; however, it had the second-highest entropy value across all ten solutions (0.583). Therefore, although our results and previous research have supported a five-profile solution, we lacked a valid reason to rule out the seven- and nine-profile solutions which, when plotted, appeared to be distinct from the five-profile solution (Figs. 2–4). Accordingly, all three solutions were retained as high-potential solutions for analysis in stage 2.

Selecting the optimal profile solution (stage 2). In stage 2, we examined the replicability of the five-, seven- and nine-profile solutions by conducting multi-group profile-similarity analysis[20] comparing each profile solution in the initial sample with a solution with the same number of profiles across nine subsamples of the initial sample. Similar to measurement invariance, profile-similarity analysis entails comparing models across two samples and progressively specifying stricter constraints by forcing sets of model parameters to be equal across samples. For example, the least restrictive set of constraints, configural similarity, involves specifying the same number of profiles across samples but allowing all other parameters (for example, mean trait levels for each profile) to be freely estimated. From least to most restrictive, we examined four sets of constraints: configural similarity (number of profiles), structural similarity (means of the indicators), dispersion similarity (variance of the indicators) and distributional similarity (proportion of sample in each profile). The same statistical indicators employed in stage 1 (that is, AIC, CAIC, BIC and SABIC) were used to compare models with successive similarity constraints. A decrease in the value of a statistical indicator with increasing constraints provides support for the higher level of similarity[20].

Inspection of the statistical values for the profile-similarity analyses revealed that the nine-profile solution did not replicate well across subsamples ($n = 9{,}909$–$10{,}142$), as it did not achieve structural or dispersion similarity with the initial sample in any comparison. That is, the values for the statistical indicators increased when moving from the less restrictive models to the more restrictive models. For example, in the first subsample, comparison of the configural similarity model to the structural similarity model resulted in increases in the AIC (1,649,997 versus 1,650,856), CAIC (1,650,521 versus 1,651,188), BIC (1,651,244 versus 1,651,646) and SABIC (1,650,827 versus 1,651,382). These results were supported by the remainder of the comparisons for determining structural similarity across the other eight subsamples, in which the values of the statistical indicators increased from the configural model to the structural model ($M\Delta_{AIC} = -945$; $M\Delta_{CAIC} = -754$; $M\Delta_{BIC} = -488$;

Table 1 | Statistical indicators for the three- to ten-profile solutions in the initial sample

	LL	SC	AIC	BIC	CAIC	SABIC	Entropy
3-profile	−718,227.976	1.314	1,436,508	1,436,753	1,436,610	1,436,670	0.567
4-profile	−716,716.868	1.305	1,433,500	1,433,810	1,433,630	1,433,705	0.592
5-profile	−715,503.986	1.416	1,431,088	1,431,464	1,431,246	1,431,337	0.559
6-profile	−714,840.736	1.651	1,429,775	1,430,218	1,429,961	1,430,068	0.565
7-profile	−714,277.359	1.561	1,428,663	1,429,171	1,428,876	1,428,999	0.558
8-profile	−713,687.416	1.530	1,427,497	1,428,071	1,427,738	1,427,877	0.560
9-profile	−713,255.624	1.578	1,426,647	1,427,287	1,426,916	1,427,071	0.583
10-profile	−712,806.168	1.706	1,425,762	1,426,468	1,426,059	1,426,229	0.568

LL, model log-likelihood; SC, scaling correction factor; AIC, Akaïke information criterion; BIC, Bayesian information criterion; CAIC, consistent AIC; SABIC, sample-size adjusted BIC.

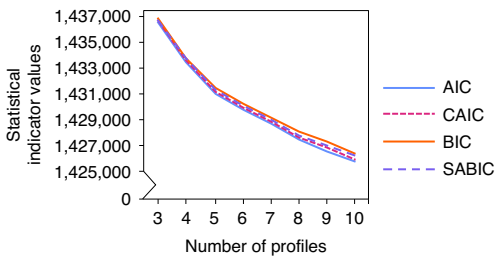

Fig. 1 | Number of profiles suggested by statistical indicators. Statistical indicators for the three- to ten-profile solutions in the initial sample (n = 90,289). To identify the correct number of profiles to extract, we examined four important statistical indices of fit: AIC[59], CAIC[60], BIC[61] and SABIC[62]. The lower the value of these fit statistics, the better the model fit. These statistical indicators are plotted and visually inspected for a point of inflection, which appears as an 'elbow' in the fit statistics on the graph. This drastic flattening of the fit statistics suggests an optimal solution. However, as our initial sample was very large, the best-fitting model could not be identified on the basis of these fit statistics, as they continued to decline as the number of profiles increased.

$M\Delta_{SABIC} = -641$, where $M\Delta$ refers to the average change in the index across all nine comparisons). Similarly, we found that the statistical indicators increased from the structural model to the dispersion model indicating that the nine-profile solution did not meet dispersion similarity in any of the comparisons with the nine subsamples: ($M\Delta_{AIC} = -1{,}040$; $M\Delta_{CAIC} = -1{,}016$; $M\Delta_{BIC} = -983$; $M\Delta_{SABIC} = -1{,}002$).

The seven-profile solution only achieved structural similarity in six of the nine comparisons. Specifically, three of the comparisons demonstrated an increase in the AIC from the configural to the structural similarity model (subsample 2 AIC = 1,650,072 versus 1,650,073; subsample 6 AIC = 1,652,094 versus 1,652,145; subsample 7 AIC = 1,653,262 versus 1,653,308). Comparisons of the remaining six subsamples demonstrated a decrease in the indicators ($M\Delta_{AIC} = 20$; $M\Delta_{CAIC} = 170$; $M\Delta_{BIC} = 402$; $M\Delta_{SABIC} = 269$). According to Morin et al., structural similarity indicates "that the nature of the profiles is similar across groups and represents a logical prerequisite to investigation of other types of similarities or differences"[20].

By contrast, the five-profile solution achieved all forms of similarity to the initial sample for all nine subsamples. Moving from the less restrictive configural model to the more restrictive structural model resulted in decreases in all statistical indicators ($M\Delta_{AIC} = 34$; $M\Delta_{CAIC} = 138$; $M\Delta_{BIC} = 303$; $M\Delta_{SABIC} = 208$), as did moving from structural to dispersion ($M\Delta_{AIC} = 9$; $M\Delta_{CAIC} = 34$; $M\Delta_{BIC} = 67$; $M\Delta_{SABIC} = 48$) and from dispersion to distributional ($M\Delta_{AIC} = 7$; $M\Delta_{CAIC} = 23$; $M\Delta_{BIC} = 45$; $M\Delta_{SABIC} = 32$). Considering these results along with those obtained in stage 1, we selected the five-profile structure as the optimal solution. Therefore, hypothesis 1a was supported.

Examining the replicability of the optimal profile solution in smaller samples (stage 3). In this stage, we used a holdout sample to create subsamples of varying sizes (n approximately 5,000–250) and conducted confirmatory multi-group LPA[27] to determine whether, and at what point, two samples of comparable size will produce different profile solutions. To this end, we used the parameters from the optimal five-profile solution in stage 1 as start values and compared the fit of models with final parameters constrained to be equal versus unconstrained across random pairings of samples of the same size. As in previous stages, decreases in the statistical indicators were treated as evidence that the constrained model fit better than the unconstrained model, suggesting that the same solution was found across samples in the comparison. Additionally, we also conducted a series of χ^2-difference tests as an additional index of replicability, so that a significant result was indicative of substantial difference between the fit of the constrained and unconstrained models, and thus a lack of replication.

The statistical indicators for the constrained models were generally better than for the unconstrained models. However, for the n = 250 samples, half of the χ^2-difference tests indicated that the unconstrained models fit better (Supplementary Table 1). These results show that, within limits, the same profile solution can be replicated across a series of increasingly small sample sizes, but they do not indicate whether the solution is the same as the optimal solution from stages 1 and 2.

To compare the solutions for the constrained models with the optimal solution, we calculated the absolute difference between each parameter estimate (that is, trait mean) for the constrained models with their corresponding parameters in the optimal solution (see Table 2 for a summary). These 'parameter-deviation' values revealed a trend whereby the parameter estimates for the constrained models differed more from the parameters of the optimal solution as sample size decreased (see Fig. 5 for a graphical representation). Specifically, the findings reveal that parameter deviations in the large samples were relatively low, suggesting that the optimal solution can be accurately recovered from relatively large samples (n approximately 5,000–750), thus providing support for hypothesis 1b. However, as sample size falls below 500, there is a radical increase in parameter deviations. Consequently, we have limited confidence that profile solutions generated using samples with fewer than 500 participants would accurately reflect the optimal solution in a general population. Indeed, it might be difficult for solutions generated from samples below an n of 500 to converge naturally on a solution obtained with any other sample, regardless of its size. This finding therefore supports hypothesis 2 and is consistent with Herzberg and Roth's claim that the lack of consistency in profile structure observed across studies might be due to the use of small samples[19].

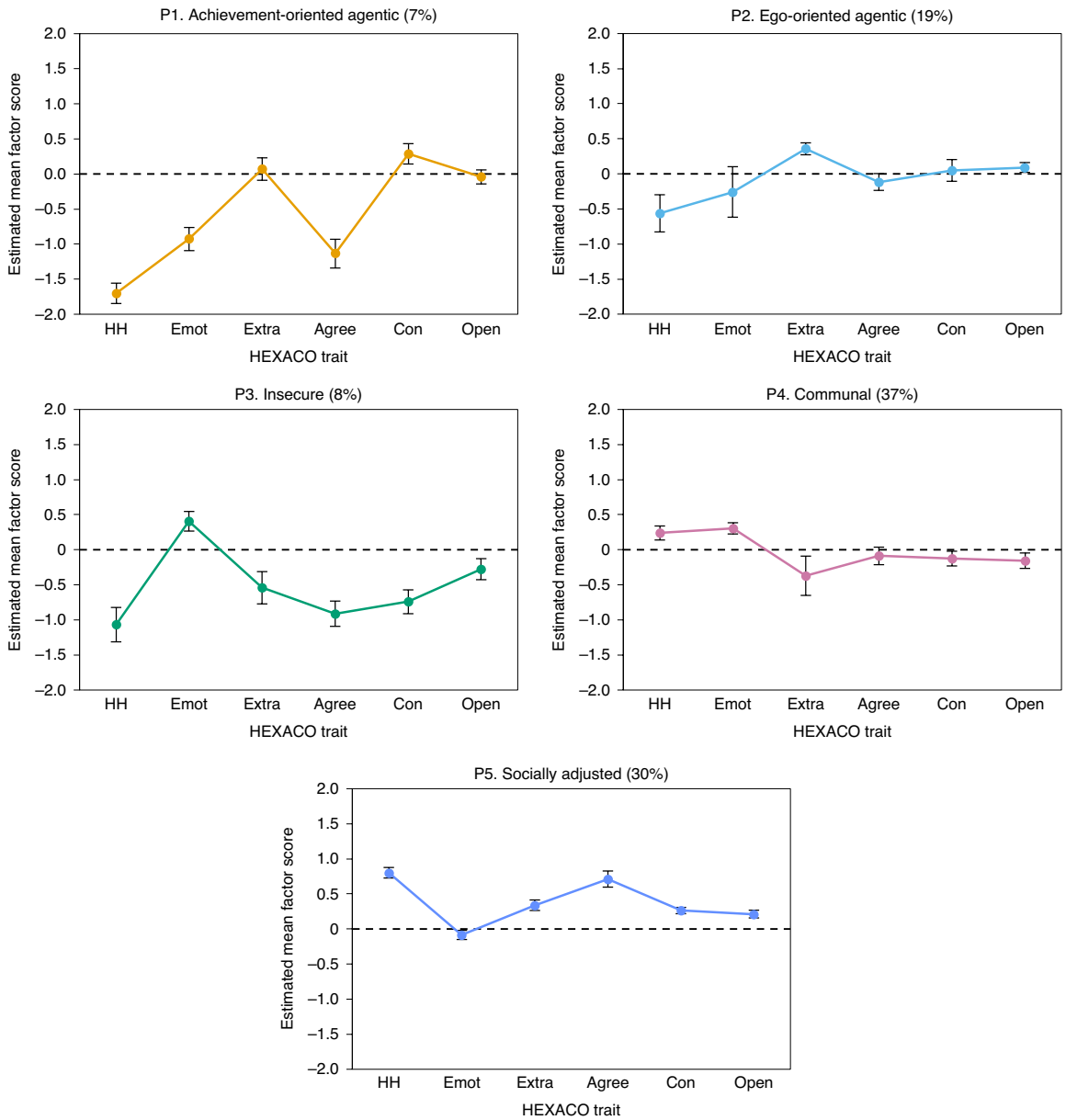

Fig. 2 | Means of the six HEXACO scales for the five-profile solution. This figure presents the five-profile solution that was generated using factor scores from the initial sample ($n = 90{,}289$). Each point on the figure identifies the estimated mean for each personality trait within the given profile. Error bars show 5 × s.e.m. Factor scores have a mean of 0 and a s.d. of 1; the estimated values for traits in each profile can therefore be interpreted in terms of s.d. above or below the mean for the sample. Estimated proportion of the sample in the profile is presented in parentheses. Each graph represents one of the five profiles in the solution. Profile 1 (P1) is the achievement-oriented agentic profile, characterized by slightly above-average extraversion, above-average conscientiousness and below-average honesty–humility, emotionality and agreeableness. Profile 2 (P2) is the ego-oriented agentic profile, characterized by above-average extraversion, slightly above-average conscientiousness and openness, slightly below-average agreeableness and below-average honesty–humility and emotionality. Profile 3 (P3) is the insecure profile, characterized by well-below-average honesty–humility, extraversion, agreeableness, conscientiousness and openness, and above-average emotionality. Profile 4 (P4) is the communal profile, characterized by above-average honesty–humility and emotionality, slightly below-average agreeableness, conscientiousness and openness, and below-average extraversion. Profile 5 (P5) is the socially adjusted profile, characterized by well-above-average honesty–humility and agreeableness, above-average extraversion, conscientiousness and openness, and slightly below-average emotionality. HH, honesty–humility; Emot, emotionality; Extra, extraversion; Agree, agreeableness; Con, conscientiousness; Open, openness.

Replication of the optimal structure using independent data (stage 4). Next, we evaluated whether the optimal profile solution could be identified in three additional independent samples of comparable size to our initial large sample (that is, $n \approx 90{,}000$). The additional data ($n = 270{,}539$) were collected via the online portal maintained by the creators of HEXACO and had similar demographic characteristics to the initial sample. The data we were provided included only cases that passed careless responding and other data-cleaning procedures. These data were analysed using the same procedures outlined in stage 1. Visual inspection of the results of these analyses indicated that the five-profile solution for each of these samples replicated the optimal solution identified in the initial

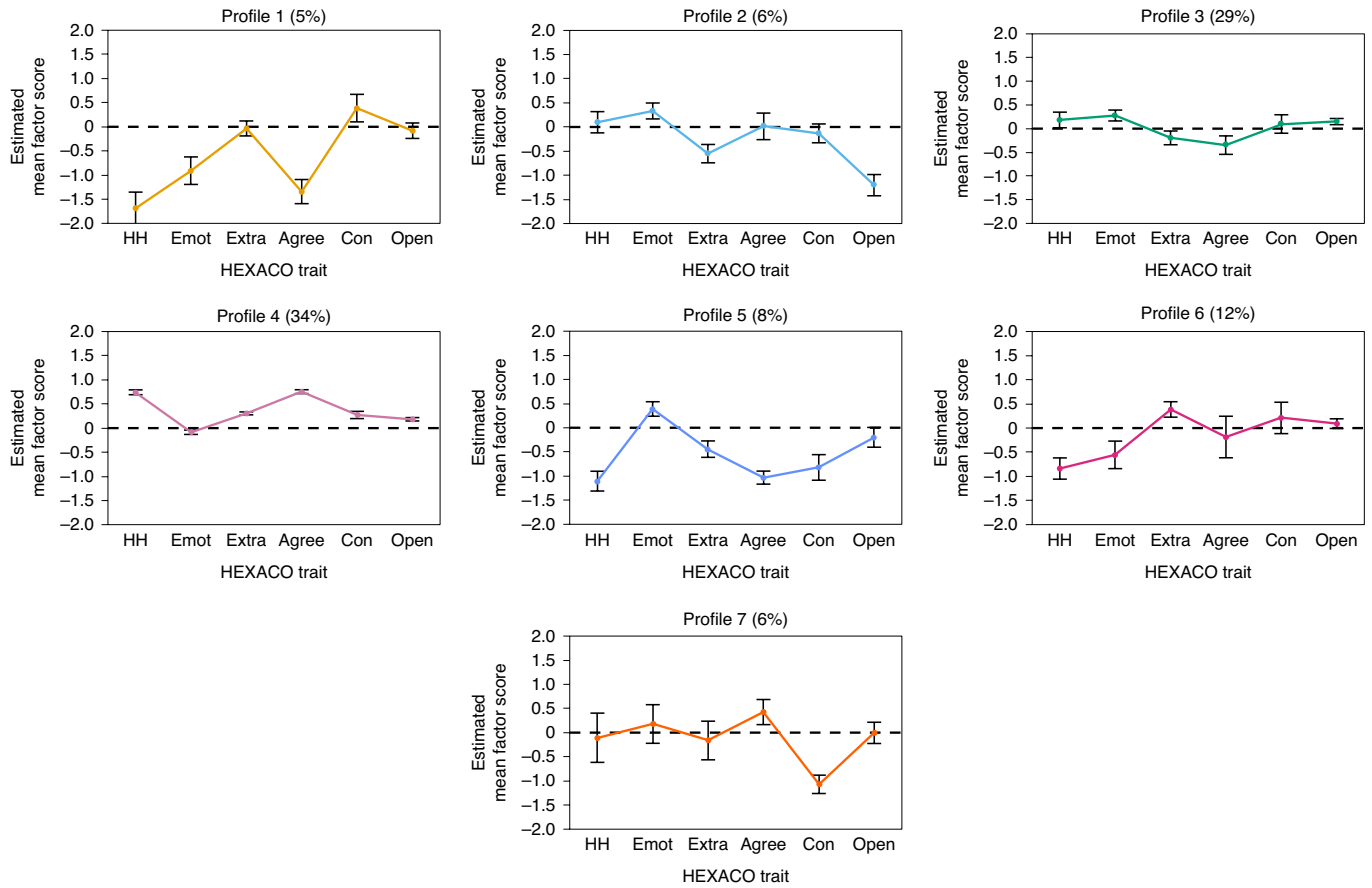

Fig. 3 | Means of the six HEXACO scales for the seven-profile solution. This figure presents the seven-profile solution that was generated using factor scores from the initial sample ($n = 90{,}289$). Each point on the figure identifies the estimated mean for each personality trait within the given profile. Error bars show 5 × s.e.m. Factor scores have a mean of 0 and a s.d. of 1; the estimated values for traits in each profile can therefore be interpreted in terms of s.d. above or below the mean for the sample. Estimated proportion of the sample in the profile is presented in parentheses.

sample (Supplementary Fig. 1a–c). Furthermore, following similar procedures as in stage 3, we calculated parameter deviations for the optimal five-profile solutions in these additional samples from the parameters for the optimal solution in the initial sample. The results of these analyses indicated that the profile solutions in each of the additional samples showed minimal deviation from the optimal profile solution. The average parameter deviation from these samples (0.08) was only slightly larger than the average deviation values found for the largest holdout samples in stage 3 ($n = 5{,}000$, deviation = 0.05; $n = 2{,}500$, deviation = 0.06). However, unlike the holdout samples, analyses for the additional samples were conducted with sample-specific factor scores. Thus, these results further attest to the validity of the optimal five-profile structure with a series of independent samples and provide support for hypothesis 1c.

Gender and age-based analyses (stage 5). At this stage, we conducted a series of exploratory analyses to investigate the potential influence of gender and age on the underlying structure of the 100-item HEXACO-PI-R and on the configuration of the profiles in the optimal solution identified in stages 1–3. Given that there has been some question about the consistency of the factor structure of the HEXACO across gender due to differences in observed scores for some traits[28], and such inconsistencies could have implications for the appropriateness of applying the same profile solution across gender, we tested for measurement invariance across gender using the initial sample. Although there has been less concern about age-related differences in factor structure, we conducted similar tests of measurement invariance across age category (that is, young adults (up to 25 yr of age), adults (26–54 yr of age) and mature adults (55 yr and older).

Analysis of measurement invariance comprises comparing models with progressively increasing constraints on multiple statistical indicators to examine the equivalence of various parameters of a measure's underlying structure across groups (for example, age and gender)[29]. For these comparisons, a decrease in the comparative fit index greater than 0.01 and an increase in root mean square error of approximation greater than 0.015 from the less restrictive to the more restrictive model is evidence that the less restrictive model is a better fit to the data and there is a lack of invariance across groups at the specified level[30].

The results of the analyses provided evidence for invariance across age and gender according to the guidelines outlined above for changes in statistical indicators, indicating that the factor structure of the HEXACO-PI-R in our data was equivalent across these categories (see Supplementary Tables 2–3 for complete results). In turn, these results suggest that the application of the same profile solution across gender and age categories was not affected by inconsistencies in factor structure across these groups.

Because of arguments that the traditional tests for invariance are not sufficiently sensitive to non-invariance with large samples[31], we conducted an additional set of exploratory analyses involving estimating independent profile solutions for the initial sample and the additional independent samples according to age and gender (Supplementary Information). Although these analyses provided

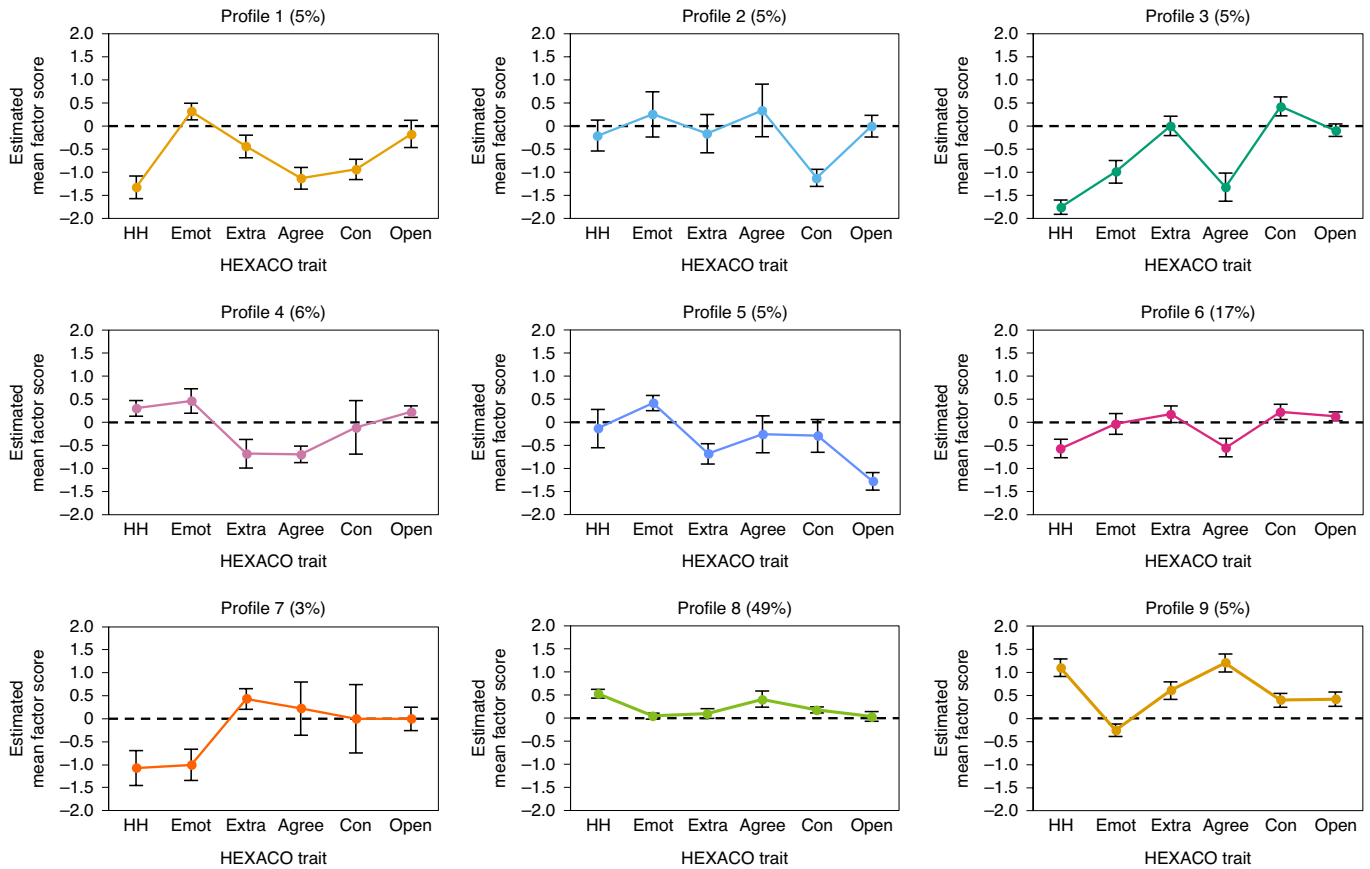

Fig. 4 | Means of the six HEXACO scales for the nine-profile solution. The nine-profile solution generated using factor scores from the initial sample ($n = 90{,}289$). Each point on the figure identifies the estimated mean for each personality trait within the given profile. Error bars show 5 × s.e.m. Factor scores have a mean of 0 and a s.d. of 1; the estimated values for traits in each profile can therefore be interpreted in terms of s.d. above or below the mean for the sample. Estimated proportion of the sample in the profile is presented in parentheses.

some evidence for differences across age groups for some profiles, the five-profile structures obtained in independent analyses for each group were very similar to the optimal five-profile solution based on visual inspection and parameter-deviation values.

Although the optimal solution appears to fit across gender and age, the aforementioned analyses do not reveal whether the gender and age groups are equally represented across profiles. Therefore, we also conducted exploratory analyses comparing the estimated relative proportion of individuals in each profile across gender and age category using data from the initial sample and the three additional samples used in stage 4 (see Table 3 for a summary and Supplementary Tables 6–8 for detailed results). Results indicated a distinct pattern in which males were relatively overrepresented in profiles 1 and 2 (4.6% and 14.0% greater representation, respectively), and females were relatively overrepresented in profiles 4 and 5 (14.7% and 5.0% greater representation, respectively). There were no substantial differences in relative proportions for profile 3. Comparisons across age groups revealed that there were relatively more young adults in profiles 1–3 than adults and mature adults (2.7%–14.7% greater representation). Comparing adults to mature adults revealed a similar pattern—there were relatively more participants in the adult category in profiles 1–4 than mature adults (3.9–10.2% greater representation). Finally, there were relatively more participants in the mature adult category in profile 5 than in any other category (22.1–35.5% greater representation). Generally, younger participants were more likely to be found in profiles 1–4 whereas the oldest participants were more likely to be found in profile 5.

Discussion

One of the primary goals of personality science is to understand the overall structure of human personality at the most general level[5]. With the variable-centred approach, this meant attempting to recover the same dimensions of personality across languages and countries[1–3,5]. In this study, we extended this work using a person-centred approach to determine whether the six HEXACO trait dimensions combine differently within subgroups of the population to reflect more holistic personality profiles.

Our results provided evidence that a five-profile solution (Fig. 2) was optimal within an initial diverse sample of more than 90,000 individuals and that a very similar solution emerged in other samples of comparable size and in smaller subsamples with $n \geq 500$. Exploratory analyses also indicated the optimal profile solution was not substantially influenced by age or gender. The fact that the optimal solution was not recoverable in samples below 500 is consistent with Herzberg and Roth's argument that the inconsistencies identified in previous profile studies might be due to the use of relatively small samples[19]. Notably, by imposing the parameters for the exploratory structural equation modelling (ESEM) and LPA analyses from our very large and diverse sample onto data from smaller samples, researchers can generate class probabilities that identify the likelihood of any individual belonging to each of the profiles in the optimal solution. Therefore, by applying these parameters in small-sample studies using the HEXACO-PI-R, researchers should be more confident in comparing findings across studies in the future. Use of these parameters also eases the research process by eliminating the need for factor-score estimation and the generation and

Table 2 | Mean absolute parameter-deviation values across holdout samples

	Mean absolute parameter deviation					
Parameter	***n*=5,000**	***n*=2,500**	***n*=1,250**	***n*=750**	***n*=500**	***n*=250**
Profile 1 means						
HH	0.122	0.127	0.138	0.193	0.542	0.847
Emot	0.035	0.114	0.143	0.206	0.375	0.848
Extra	0.029	0.096	0.149	0.094	0.312	0.098
Agree	0.125	0.130	0.227	0.142	0.263	0.656
Con	0.006	0.026	0.085	0.057	0.349	0.438
Open	0.036	0.068	0.105	0.182	0.163	0.305
Profile 2 means						
HH	0.101	0.099	0.189	0.091	0.444	0.575
Emot	0.174	0.160	0.173	0.178	0.422	0.525
Extra	0.026	0.042	0.059	0.039	0.304	0.250
Agree	0.087	0.107	0.279	0.185	0.362	0.641
Con	0.005	0.023	0.242	0.135	0.209	0.558
Open	0.012	0.029	0.057	0.039	0.145	0.191
Profile 3 means						
HH	0.009	0.061	0.150	0.071	0.177	0.545
Emot	0.017	0.039	0.145	0.062	0.351	0.377
Extra	0.102	0.114	0.093	0.131	0.093	0.368
Agree	0.012	0.049	0.141	0.124	0.158	0.199
Con	0.020	0.055	0.141	0.164	0.096	0.318
Open	0.008	0.086	0.207	0.173	0.346	0.286
Profile 4 means						
HH	0.006	0.025	0.146	0.073	0.339	0.694
Emot	0.024	0.029	0.034	0.085	0.160	0.520
Extra	0.122	0.110	0.144	0.108	0.399	0.595
Agree	0.014	0.024	0.100	0.095	0.186	0.556
Con	0.038	0.038	0.078	0.056	0.106	0.406
Open	0.036	0.040	0.044	0.047	0.222	0.412
Profile 5 means						
HH	0.044	0.041	0.063	0.104	0.112	0.111
Emot	0.048	0.049	0.029	0.032	0.138	0.081
Extra	0.045	0.037	0.096	0.085	0.156	0.230
Agree	0.063	0.066	0.033	0.085	0.090	0.147
Con	0.005	0.024	0.065	0.043	0.040	0.139
Open	0.033	0.036	0.061	0.082	0.120	0.143
Comparisons	1	6	6	6	6	6

Comparisons is the number of sample pairings included in mean parameter deviations; *n* is the number of participants in each sample in the model.

comparison of multiple profile solutions using LPA. We have provided syntax files containing the ESEM factor structure we obtained and the parameter estimates from our five-profile solution in the Supplementary Software.

It is important to note that, as an analytic tool, LPA is probabilistic in nature, which means that the results of profile analysis yield an estimate of participants' likelihood of presenting each profile within the solution[32,33]. Consequently, contrary to the concerns of some critics of the profile approach, we are not claiming to be "carving nature at its joints"[11,34] by arguing that the five profiles reflect discrete categories of people. Rather, we argue that these profiles exemplify a set of cognitive, behavioural and affective characteristics to which any individual will show some degree of similarity. Thus, just as scores from the HEXACO (and FFM) scales indicate the degree to which specific trait definitions characterize an individual, the posterior probabilities generated in LPA reflect the likelihood that an individual has a trait profile resembling the prototypical trait configurations.

At this point, it would be premature to assign definitive labels to the profiles, but it is nevertheless interesting to compare the profiles to those obtained previously as a way to begin the labelling process. Recall that many earlier profile studies involving the FFM[11–13] and HEXACO[18] used labels derived from Block's self-regulation theory[14,15]. Of the profiles obtained, only the profiles reflecting high scores on all traits (positively keyed) emerged consistently. Using Block's terminology, this profile is commonly labelled 'resilient' to

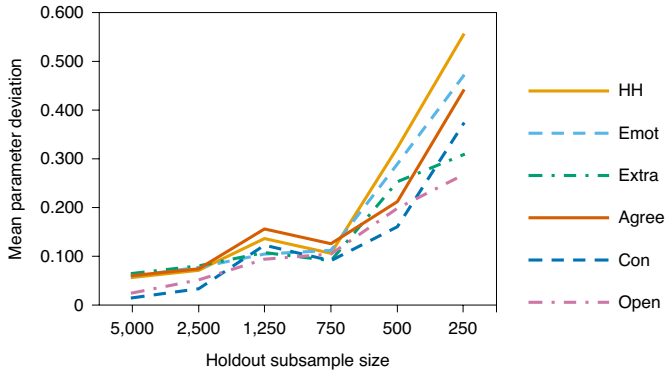

Fig. 5 | Mean absolute values of parameter deviations from the optimal five-profile solution. Mean absolute deviation between each parameter estimate (that is, trait mean) for the constrained models with their corresponding parameters in the optimal solution (Table 2) on a per-trait basis. The graph depicts the relationship between absolute parameter deviation and sample size, demonstrating that, as sample size drops below 500, the average absolute parameter deviation begins to exceed 0.50. Because the parameter estimates of our models are calculated using factor scores which have a mean of 0 and a s.d. of 1. For example, an absolute parameter-deviation value of 0.50 indicates that the parameter from a given model diverges from the corresponding parameter from the optimal solution by half a standard deviation.

Table 3 | Average differences in estimated relative proportions in each profile according to gender and age category

Profile	$M\Delta_{m\text{-}f}$	$M\Delta_{ya\text{-}ad}$	$M\Delta_{ya\text{-}ma}$	$M\Delta_{ad\text{-}ma}$
P1. Achievement-oriented agentic	+4.6%	+2.7%	+5.3%	+10.2%
P2. Ego-oriented agentic	+14.0%	+5.6%	+14.6%	+9.0%
P3. Insecure	+0.4%	+4.5%	+8.3%	+3.9%
P4. Communal	−14.7%	+0.7%	+7.3%	+6.7%
P5. Socially adjusted	−5.0%	−13.4%	−35.5%	−22.1%

$M\Delta_{m\text{-}f}$ is the average difference in proportion (as a percentage) between males and females in each profile across samples; similarly, $M\Delta_{ya\text{-}ad}$, $M\Delta_{ya\text{-}ma}$ and $M\Delta_{ad\text{-}ma}$ are the average difference in proportion (as a percentage) of young adults and adults, young adults and mature adults, and adults and mature adults, respectively, between participants in each profile across samples. Positive values indicate a greater relative proportion of participants in the first category for each comparison.

describe individuals who can modulate their natural control tendencies (that is, free expression or inhibition). We found a similar profile (profile 5) in our study. However, there are two reasons we are reluctant to use this label. First, the mere fact that individuals with this profile have high scores on all the HEXACO traits except emotionality does not provide evidence that they are able to modulate their control tendencies. Second, the concept of resilience has become the focus of a large body of theoretical and empirical work[35–38] that goes well beyond its conceptualization in Block's theory. Consequently, its continued use as a personality profile label could be confusing. Instead, we used the label 'socially adjusted' for theoretical reasons described below.

The remaining profiles found in previous studies have often been inconsistent (despite being given similar labels such as overcontrolled and undercontrolled). Thus, we avoided the use of these labels. Indeed, given the differences between the FFM and HEXACO models of personality[1], comparisons with previous FFM studies are not particularly meaningful. Comparisons with previous HEXACO studies[16,18] are more appropriate, albeit limited. Although Isler et al.[18] purportedly identified HEXACO profiles, they did so by adding honesty–humility-relevant items to the international personality item pool (IPIP) measure of the Big Five. Consequently, their data do not reflect other differences between the HEXACO and FFM (for example, emotionality versus emotional stability) that can have implications for interpretation. Only the Daljeet et al. profiles obtained using the HEXACO-60[16] provide a legitimate basis for comparison.

Daljeet et al. observed a consistent set of five profiles across two samples of university students[16]; some of our profiles resemble these five profiles, the most similar being our profile 5 and Daljeet's profile 1 (all traits at or above average). Daljeet et al. used the HEXACO facet traits to provide descriptive labels. In this case, they applied the label 'socially considerate.' Although the remaining profiles they obtained have some resemblance to those obtained in the present study, the differences are sufficient to make us reluctant to adopt their labelling scheme. The consistency in the profile structure observed by Daljeet et al. across samples suggest that their profiles are meaningful for a student sample but, as the authors acknowledge, they do not necessarily reflect the profiles likely to characterize people in general. Our profiles are based on a much larger and more diverse sample, demonstrate a high degree of consistency, and stem from responses to a more extensive measure. Thus, rather than adopting the labels from Daljeet et al., we turned to consideration of HEXACO theory and other established theories of personality as a starting point for a labelling scheme.

According to Ashton and Lee[39], the HEXACO traits can be divided into two distinct clusters. The first cluster includes honesty–humility, agreeableness and emotionality, all of which have an altruistic social focus. The second cluster includes extraversion, conscientiousness and openness to experience which, in contrast to the first cluster, have more of a task focus. Ashton and Lee argued that these trait clusters have parallels in various theories of personality and human functioning and may have unique biological bases. For example, the altruistic social cluster includes traits reflecting a tendency toward reciprocal caring that was essential for nurturing and protection of the young and is most prevalent in females. By contrast, traits in the task-focused cluster foster engagement and accomplishment in social- (extraversion), task- (conscientiousness) and idea-related (openness to experience) endeavours. Therefore, it is interesting to note that three of the five profiles obtained in our study reflect above-average scores on traits in one cluster and below-average scores on traits in the other cluster. That is, profile 1 includes above-average scores on conscientiousness and well-below-average scores on honesty–humility, agreeableness and emotionality. Profile 2 is characterized by an above-average score on extraversion and below-average scores on honesty–humility, emotionality and agreeableness. By contrast, profile 4 is characterized by above-average scores on honesty–humility and emotionality, with below-average scores on the remaining traits, including the task-focused cluster (particularly extraversion). Only profile 5 is characterized by elevation in both social-focused and task-focused traits. The one remaining profile, profile 3, with its above-average score on emotionality and below-average scores on the remaining social- and task-focused traits, is more difficult to classify according to HEXACO theory. It appears to be the inverse of profile 5.

There are other theories that might also help to explain the varying trait configurations. For example, the contrast between profiles 1 and 2 and profile 4 could reflect the distinction between the agentic and communal orientations, respectively, first identified by Balkan[40] and subsequently discussed as basic dimensions of personality[41,42]. By contrast, profile 5 might reflect a balance between agentic and communal concerns. Profile 3 is again somewhat difficult to classify, but might reflect the absence of both agentic and communal qualities. Of note, the exploratory analyses we conducted examining gender and age revealed distributional differences consistent with

both HEXACO and agency–communion theories. That is, males tended to be overrepresented in the two agentic profiles (profiles 1 and 2) and females were overrepresented in profiles with elevated communal traits (profiles 4 and 5). Similarly, representation in the agentic profiles generally declined with age, whereas representation in the more communal profiles increased with age.

Another theory that might complement the emphasis on evolutionary biology in HEXACO theory is Bowlby's attachment theory[43]. Here the emphasis is on the influence of early experiences with caregivers that can shape self-concept and social interaction styles. Individuals who develop a secure attachment style resulting from emotionally supportive and protective caregivers might be most likely to have a personality reflecting the trait configuration in profile 5. Those who develop less secure styles (for example, anxious or avoidant), might be more likely to have personalities reflecting high levels of emotionality as in profile 3. Notably, we observed little difference in the proportions of males and females in profile 3, leaving open the possibility that membership in this profile is more a function of socialization than biology.

A more detailed discussion of the many personality and related theories that could help to explain the emergence of a core set of HEXACO trait profiles goes beyond our current objectives. Nevertheless, the fact that our findings reveal trait configurations that are consistent with HEXACO, agency–communion and attachment theories contributes further to our confidence in the meaningfulness of the profiles and provides a basis for the assignment of tentative labels. Profile 1 appears to reflect a self-focused achievement orientation and is therefore labelled achievement-oriented agentic. Profile 2 reflects an agentic orientation focusing on self within a social context and is labelled ego-oriented agentic. Profile 3 reflects a high degree of emotionality that supersedes, and might interfere with, task accomplishment and social relationships. We labelled this profile insecure. Profile 4 appears to present a communal as opposed to task orientation and is labelled communal. Finally, Profile 5 seems to reflect a healthy balance of agentic and communal concerns with below-average emotionality and is therefore labelled socially adjusted.

Although our labels and the implication they have for interpretation must be considered tentative at this time, our findings set the stage for future research in which probabilities of profile membership can be linked to measures of attitudes, values and behaviours as a way of establishing the nature of the underlying profiles. These findings and the ability to reproduce a common profile structure even in relatively small samples also afford the opportunity to use a person-centred approach to revisit issues addressed in previous variable-centred research. Personality has long been used to understand and predict important phenomena such as job performance[44,45], leadership behaviours[46], academic performance[47], academic dishonesty[48], happiness[49], political attitudes[50] and communication styles[51]. However, this research focuses largely on the independent or additive contributions of individual traits. Using a person-centred approach in the future can provide several advantages for the investigation of how personality traits combine to influence these phenomena.

A first advantage is that person-centred analytic techniques are better suited than variable-centred techniques to detect complex trait interactions. Achieving the power to detect even simple interactions in variable-centred analyses (for example, moderated multiple regression) requires large samples[52]. Further, higher-level interactions such as those required to distinguish members of any of the profiles identified in our study (that is, six-way interactions) would be almost impossible to detect, let alone interpret. By contrast, these complex interactions emerge in comparisons across profiles in the person-centred approach and can be relatively straightforward to explain.

A related advantage of the person-centred approach is that it acknowledges that a given trait might manifest differently depending on how it combines with other traits in a profile[16]. For example, comparison of profiles 1 and 5 in the optimal five-profile solution reveals similar levels of conscientiousness combined with very different patterns among other traits (Fig. 2). Most notably, honesty–humility and agreeableness are very low in profile 1 but quite strong in profile 5. Consequently, we might expect conscientiousness to be expressed more in terms of self-interest (for example, achievement striving) in profile 1, and in terms of social interest (for example, fulfilling obligations) in profile 5. Likewise, extraversion scores are similar in profiles 2 and 5 but, again, the remaining traits reflect a quite different pattern. When combined with strong honesty–humility and agreeableness in profile 5, extraversion might be expressed in an outgoing and congenial manner. By contrast, as the dominant trait in profile 2, it might manifest as narcissism or Machiavellianism. Such interpretations are merely speculation at this point, and actual interpretation of the profiles requires further research. However, if the traits do operate in concert it raises questions about the meaningfulness of findings pertaining to the traits in isolation.

A third advantage of using a person-centred method such as LPA is that it calculates the probability that an individual belongs to each of the identified profiles and allows for easy extraction of these values. These posterior probabilities can then be used as continuous variables in the investigation of potential antecedents or outcomes of profile membership. This is an important advantage over variable-centred strategies where, even if trait interactions are detected, there is no way to 'quantify' the interaction to determine where an individual is situated within the interaction (that is, indicate the specific combination of traits being manifest). Without quantification, the investigation necessarily stops at the detection of the interaction and assumes that the reason for, and nature of, the effects is consistent within the entire sample.

A limitation of our study is that, despite comprising almost 400,000 English-speaking participants from around the world, our total sample is not necessarily representative of the entire human population. Moreover, it is possible that even larger and more representative samples, such as those used in Gerlach et al.[17], could yield different results when examined using our procedures. Even so, a major contribution of this study is the introduction of an analytic strategy to be used with these data. That is, the methodology we advance here is generalizable, even if our findings are not.

Another limitation of our study is that the parameter estimates for our profiles were derived using the 100-item version of the HEXACO-PI-R. Although this version of the instrument has desirable psychometric properties, there are other inventories that also measure the HEXACO dimensions. Accordingly, it would be beneficial for future researchers to apply the procedures outlined in our study to develop a similar set of parameter estimates suitable for use with other measures including IPIP-HEXACO[53], the 60-item version of the HEXACO[54], and the Brief HEXACO Inventory[55]. Determining whether the same five profiles can be identified using different measures of the HEXACO could have implications for the meaningfulness and validity of the profiles we identified. Moreover, it would allow for a broader set of existing datasets to be re-examined using a person-centred approach, expanding the range of constructs that can be investigated to characterize individuals associated with each profile.

Finally, our study was also limited to the identification of HEXACO trait profiles and their cross-sample consistency, but did not address the meaning, development, temporal stability or implications of the profiles empirically. We used selected personality theories to offer tentative explanations for the emergence of the five profiles and encourage future research to test these and other theory-based propositions. The ultimate utility of the person-centred (profile) approach to the investigation of the HEXACO traits will rest on the results of programmatic research designed to gain greater insight into the interpretation, development, and temporal

stability of the profiles as well as their contribution to understanding of behaviour beyond the individual traits.

In summary, the results of our study provide strong evidence that there are at least five dominant profiles in the HEXACO space. We tested the replicability of these profiles across a variety of large and small samples drawn from a 'population-like' sample. Exploratory analyses revealed that distributional differences across gender and age in these profiles also appear to be in line with some personality theories, further supporting their validity. Although the same five profiles were easily recovered from our large samples ($n \geq 500$–10, 000), our results clearly showed that as sample size decreased it became increasingly difficult to accurately recover the five-profile solution we found was optimal in the population-like sample. We encourage other researchers to use the results and methods outlined in our study as a stimulus for future person-centred research on the structure, function, replicability and utility of profiles in the personality space.

Methods

Participants and procedure. The initial data used in the present study were collected between 2014–2015 via an online portal maintained by the creators of the HEXACO[56]. For present purposes, we used only the data from the 100,318 (of 104,467) who completed all 100 items from the English version of the HEXACO-PI-R and correctly responded to three attentiveness-check items. Collection of these data was approved by the research ethics board at the University of Calgary. Participants provided informed consent prior to participating in the study and were not compensated for their participation. Additional details regarding data collection, data cleansing and the psychometric properties of the HEXACO-PI-R are reported by Lee and Ashton[56].

Women and men made up 48.4% and 50.2% of the sample, respectively (the remainder did not indicate gender), and the average age of participants was 37.1 yr (s.d. = 14.1 yr). Most participants were of European descent (64.7%), with others identifying as African (1.4%), Arab and/or West Asian (1.3%), Australian Aboriginal (.2%), East Asian (3.2%), Latin American (2.6%), North or South American Aboriginal or Native (1.0%), 'other' (7.1%), Pacific Islander (.2%), South Asian (4.3%), Southeast Asian (2.9%) or 'two or more of the other options' (4.0%); the remainder did not respond or chose 'prefer not to answer.' A large percentage of the sample indicated that they completed university/college (41.6%), followed by graduate or professional school (32.8%), or high school (19.2%).

The additional data ($n = 270{,}539$) used in stage 4 of our analyses originated in subsequent collections (performed in 2015–2018) on the online portal maintained by the creators of the HEXACO and underwent similar data-cleaning procedures as the initial sample. The data we were provided included only cases that passed careless responding and other data-cleaning procedures. There were minor demographic differences between this and the initial sample. This sample included approximately 10% more men (men, 60.0%; women, 39.0%), was comprised of younger individuals on average (mean age = 27.6 yr; s.d. = 11.1 yr), with approximately 15% fewer participants of European descent (49.8%). A larger portion of the replication sample reported having a high-school education (32.4%) and a smaller percentage indicated that they completed a graduate or professional education (16.8%).

No statistical methods were used to pre-determine sample sizes because the present research used the entirety of the archival data made available to the authors. Additionally, our sample sizes are many times larger than those reported in previous publications examining profiles in the HEXACO personality space[16,18].

Measures. *Personality.* The HEXACO-PI-R contains 16 items assessing each of the six broad HEXACO traits: honesty–humility (for example, "I would never accept a bribe, even if it were very large"), emotionality (for example, "I feel like crying when I see other people crying"), extraversion (for example, "I enjoy having lots of people around to talk with"), agreeableness (for example, "I generally accept people's faults without complaining about them"), conscientiousness (for example, "I often check my work over repeatedly to find any mistakes"), and openness to experience (for example, "I'm interested in learning about the history and politics of other countries"). It also includes four items assessing the interstitial facet of altruism (for example, "I try to give generously to those in need"). The items addressing altruism were included in the honesty–humility, emotionality and agreeableness scales, dividing their loadings across these three factors in all analyses. All responses were made on a five-point Likert-type scale from 1 (strongly disagree) to 5 (strongly agree).

Analyses. In preparation for analyses of data from the initial sample, we created two subsamples by randomly assigning 10 per cent of the total sample to a holdout group ($n = 10{,}029$), with the remainder ($n = 90{,}289$) constituting the initial sample. Analyses with these data were then conducted in three stages as described below. A fourth stage of analysis was conducted to examine replicability using data independent from our holdout and initial samples ($n = 270$ and $n = 539$, respectively). The fifth stage was conducted for exploratory purposes and examined measurement invariance across gender and age in the initial sample and included descriptive analyses of estimated profile membership on the basis of gender and age for the large samples used in stage 1 and stage 4.

All analyses were conducted in Mplus 7.0 using robust maximum-likelihood estimation. Robust maximum-likelihood estimation calculates fit indices and standard errors robust to violations of normality assumptions, although violations of these assumptions were not formality tested.

Identifying high-potential profile solutions (stage 1). In stage 1, we used LPA to generate solutions ranging from three to ten profiles. On the basis of previous research, we expected that the optimal solution would fall within this range. The LPA were conducted using factor scores generated using ESEM analyses with the HEXACO-PI-R items. We used factor scores rather than raw-score composites because they help provide some control for measurement error under conditions like the present where the use of latent factors is too computationally demanding[20]. We used ESEM rather than confirmatory factor analysis to generate the factor scores because confirmatory factor analysis has been criticized for requiring that the loadings of all items in a model be constrained to zero on all but their intended factor[57]. The assumption of zero cross-loadings is unrealistic because it is unlikely that any one item reflects only a single factor, especially when there are multiple factors in a model that might have some degree of overlap. ESEM allows for modest cross-loadings of items on factors other than their intended factor using target rotation procedures. Target rotation allows all items to load on all factors but specifies that any cross-loadings be held as close to zero as possible. Accordingly, factor scores extracted from ESEM might also allow for the factor scores extracted from a model to better reflect the theoretical structure because it leverages the variance in all indicators accounted for by a factor, including variance in items that were not intentionally designed to reflect a factor[58].

LPA is an iterative procedure that allows for the identification of subgroups within a sample based on their similarity on a set of variables, in this case the factor scores reflecting the HEXACO traits. Unlike other person-centred analyses (that is, cluster analysis), LPA is model-based and provides multiple statistical indicators that can be used as a guide to select the optimal profile solution for the data. To identify high-potential solutions, we examined various statistical indicators including AIC[59], CAIC[60], BIC[61] and SABIC[62]. A decrease in these statistical indicators marks an improvement in fit; however, because these indicators are based on tests of significance, they can be heavily influenced by sample size[31]. In turn, with large sample sizes, improvement in these statistics should be examined using elbow plots[20,25,63], in which a flattening of the slopes for these statistics suggests that an optimal solution has been identified. We also examined the entropy[26] for each solution. Entropy provides an indication of classification accuracy, ranging from 0 to 1, with higher values indicating greater accuracy. Additionally, we examined what percentage of the sample was associated with each profile within a given profile solution. A solution was considered inadequate if any profile within the solution was estimated to contain less than 5% of the sample as it might be too small to be robust[26]. Following the procedure outlined in previous work[16,26], we considered all available information when identifying high-potential solutions. In addition to statistical indicators and profile size, we also considered the distinctiveness of the profiles within a solution based on their shape and elevation[24].

Selecting the optimal profile solution (stage 2). Analyses in this stage examined the replicability of any solution in stage 1 that was identified as having high potential as the optimal solution. For stage 2, the initial sample was randomly split into nine subsamples of approximately 10,000 participants each. The procedures outlined in stage 1 were followed for each of the nine subsamples; ESEM was used to generate factor scores for the HEXACO model and the resulting factor scores were then used as input for LPA to estimate solutions with the same number of profiles as the high-potential solutions.

Next, we conducted multi-group profile-similarity analysis as described in Morin et al.[20] to determine which high-potential solutions replicated consistently and accurately across the nine subsamples when compared with the initial sample. That is, we conducted a set of similarity analyses between our initial sample, and each of our nine subsamples, and repeated this process for each of the high-potential solutions identified in stage 1. Profile-similarity analysis is akin to analysis of measurement invariance[64] in that it is a sequential process of specifying equality constraints to examine progressively stricter forms of invariance, or in our case, profile similarity. There are six forms of profile similarity, of which only the first four are relevant to our study[20], and decreases in statistical indicators from one set of constraints to the next indicates support for the model with stricter constraints. The first form of similarity, configural similarity, identifies whether there are the same number of profiles in both groups. Structural similarity identifies whether the means of the indicators within a profile are equal across groups. Dispersion similarity identifies whether the variance of the indicators is equal across groups. Distributional similarity—the strictest form of similarity—identifies whether the proportion of individuals in each profile is the same across groups[20]. The solution that most frequently achieved the highest degree of similarity across the nine comparisons was considered to have the greatest degree of replicability (consistency and accuracy) and was selected as the optimal solution for the initial sample.

Examining the replicability of the optimal profile solution in smaller samples (stage 3). The initial goal of stage 3 was to examine the recoverability of the optimal solution in increasingly smaller portions of the holdout sample that simulated sample sizes commonly found in trait profile research. By evaluating the relationship between the recoverability of the optimal solution and sample size, we sought to establish some guidelines for future research and identify at what point a sample becomes too small to accurately recover a known profile solution. To achieve this objective, the holdout sample was first resampled six times with replacement resulting in two samples of 5,000, four samples of 2,500, eight samples of 1,250, 13 samples of 750, 20 of 500, and 40 samples of 250. Each sample was entirely independent of other samples in the same sample-size category (that is, no data from participants in the first sample of 5,000 were included in the second sample of 5,000, but all participants were eligible for selection when creating the various samples of 2,500 participants).

To maintain consistency and to avoid proliferation of analyses we conducted six sets of paired comparisons for each sample size, except for $n = 5{,}000$, where only a single comparison was possible. Due to the complexity and associated computational intensity of the planned analyses, we opted to randomly select four holdout samples for each sample size that generated more than four samples (for example, samples of $n = 1{,}250$) for subsequent analyses. The use of four samples allowed us to examine the stability and replicability of our results in consideration of the computational power and time needed to conduct the analyses with all possible pairings of holdout samples at each sample size. For example, using all 13 samples of $n = 750$ would lead to analysis of 78 sample pairings. In comparison, using four of these samples requires an analysis of six sample pairings.

For each of the six categories of holdout samples, we followed the same basic analytic procedures as in stages 1 and 2, with one exception. With increasingly small samples, the likelihood of sampling error increases. This error could influence the factor structure of the HEXACO as well as the factor scores that serve as input to the LPA. Because our aim was to examine the recoverability of the optimal profile solution, not the recoverability of the factor scores, we chose to control for the potential impact of sample size and the associated error by fixing the parameters for the ESEM of the holdout samples to the values produced by the ESEM of the initial sample.

Using factor scores as input, we conducted confirmatory multi-group LPA based on Schmiege, Masyn, and Bryan[27] to determine whether the optimal solution identified in stage 1 could be recovered using increasingly small samples. We conducted the confirmatory multi-group LPA by selecting up to four samples from each sample-size category and comparing five-profile models with and without parameters being constrained to equality. In this procedure, we used the parameter estimates from the optimal solution as start values for both the constrained and unconstrained models. If the constrained model demonstrates equal or superior fit in comparison to the unconstrained model, it can be interpreted as evidence that the same profile solution is an optimal fit for both samples.

Additionally, to determine the degree to which the solutions for the constrained models differed from the optimal solution, and to track the trajectory of these differences across sample-size categories, we computed a series of parameter-deviation values. These values are the absolute difference between each parameter estimate (that is, trait mean) for the constrained model and the relevant parameter for the optimal solution. Note that factor scores are used as input for LPA and have a mean of zero and a s.d of one. Thus, an average parameter-deviation value of 0.50 for the solutions in one sample-size category indicates that the parameters for those solutions diverge from the parameters of the optimal solution by $0.5 \times$ s.d on average.

Replication of the optimal structure using independent data (stage 4). To determine whether the optimal five-profile structure obtained with our initial sample would replicate in independent samples of comparable size, we used data from subsequent online administrations of the HEXACO-PI-R ($n = 270{,}539$). With these data, we created three subsamples of approximately the same size as the initial sample ($n \approx 90{,}000$). We conducted freely estimated ESEMs on each of the three samples and used the resulting factor scores as input for LPA. We extracted a five-profile solution from each sample. As in stage 3, we calculated parameter-deviation values for each of the three additional samples with respect to the optimal profile solution to examine replicability.

Gender and age-based analyses (stage 5). In this final stage, we conducted exploratory analyses comparing the optimal five-profile structure across gender and age category. We conducted these analyses using data for the initial sample and the independent samples used in stage 4. We excluded data from participants who did not indicate their gender (approximately one per cent) or provide their age (approximately two per cent) from the relevant comparisons. Complete details on the number of participants excluded from the analyses for each sample across gender and age are reported in Supplementary Table 6. For the age-based analyses, we assigned respondents to three broad age categories: young adults (up to 25 yr of age), adults (26 to 54 yr of age) and mature adults (55+ yr of age).

To examine whether there were inconsistencies in the factor structure of the 100-item HEXACO-PI-R across gender and age that could impact the appropriateness of applying the same profile solution across these groups, we conducted a sequential set of multi-group ESEM analyses[29] to test for measurement invariance. These tests involved comparisons of models with progressively increasing constraints. The first test (configural invariance) involved the comparison of six-factor solutions across category without any parameter constraints. Assuming that this model fit well, we compared models with equality constraints placed on factor loadings (metric invariance), item intercepts (scalar invariance), uniquenesses (strict invariance), and both variance–covariance and latent means (complete invariance). In comparing models, we relied on guidelines provided by Chen et al.[30], suggesting that a decrease in comparative fit index greater than 0.01 and an increase in root mean square error of approximation greater than 0.015 from the previous model is evidence that the less restrictive model is a better fit to the data.

The results of the invariance analyses indicated it was appropriate to apply a single profile solution across all gender and age groups (see Results for more details). However, applying the optimal profile structure to all groups does not necessarily indicate that these groups are equally represented across profiles. Therefore, we also conducted exploratory analyses to examine whether representation in the profiles of the optimal five-profile solution differed across age and gender. These comparisons of estimated relative membership in each profile across gender and age category were conducted using data for the initial sample and the independent samples used in stage 4. To compare estimated relative proportions, individuals were first assigned to the profile for which they had the highest probability of membership as indicated by posterior class probabilities. We then determined the proportion of individuals from each group (that is, gender or age category) assigned to each profile and compared the proportions across relevant groups. For example, we compared the proportion of all females assigned to profile 1 to the proportion of males assigned to the same profile. This allowed us to account for difference in both profile size and in the number of males and females within the full sample.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study were used with permission from K. Lee and M. Ashton[56] and are not publicly available. Requests for access to the data should be directed to K. Lee and M. Ashton.

Code availability

Syntax required for the use of the final factor structure and optimal profile solution identified in this research are available in the Supplementary Software for this paper and online in the form of Mplus.inp files (https://osf.io/qdv8z/). Syntax is provided for use in Mplus 7.0 or above and annotations describe the functions of different sections of the code.

Received: 7 March 2019; Accepted: 5 March 2020;
Published online: 6 April 2020

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Acknowledgements

The authors thank K. Lee and M. Ashton for sharing the data used in this research and for their comments on an earlier version of this manuscript. This research was supported by funding to J.P.M. from the Social Sciences Faculty Research Development Grant (2018–19) at The University of Western Ontario. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Author contributions

K.N.D., J.P.M. and J.A.E. conceptualized the research, interpreted results and wrote the manuscript. J.A.E. and K.N.D. designed the methods and performed the analyses. J.A.E. and K.N.D. contributed equally to this article and both should be considered first authors.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41562-020-0853-6.

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Peer review information Primary Handling Editor: Marike Schiffer.

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Last updated by author(s): Feb 29, 2020

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Software and code

Policy information about availability of computer code

Data collection	Data were collected in another study (Lee & Ashton, 2018) using an online portal (www.hexaco.org). Participants provided access to their anonymized responses for use in research in exchange for a report on their personality according to the HEXACO model of personality.
Data analysis	Analyses were conducted using Mplus 7.0. Syntax for the replication of the final factor structure and optimal profile solution are provided in the Supplementary Information with annotations as well as an online repository at https://osf.io/qdv8z/.

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The data that support the findings of this study were used with permission from Dr. Kibeom Lee and Dr. Michael Ashton (Lee & Ashton, 2018), and so are not publicly available. Data are however available from the authors upon reasonable request and with permission from Dr. Lee and Dr. Ashton.

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Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This research uses a large dataset of individual responses to the HEXACO-100 measure of personality to identify subgroups that reflect combinations (profiles) of the HEXACO traits present in the sample. We use various techniques using subsamples and holdout samples of the dataset to examine the structure and replicability of multiple profile solutions and identify the optimal solution to the data. We also conduct exploratory analyses to determine the estimated proportions of individuals in each profile in the optimal profile solution according to gender and age.
Research sample	The dataset used in this sample was collected through an online portal by the creators of the HEXACO model of personality (Lee & Ashton, 2018; www.hexaco.org). The initial dataset is large (n = 100,318) and contains responses from individuals with a wide range of ages, ethnicities, and gender. As such, we expected that it provided a reasonable estimation of the population of HEXACO test takers allowing us to address concerns with previous research in this area conducted with small (N < 500) and non-representative samples (e.g., only university students). An additional independent dataset (n = 270,539) was used to examine the replicability of the solution identified in the initial sample. Participants responded to the complete measure of the HEXACO model, the HEXACO-100, assessing individuals on the six HEXACO personality traits (Honesty-Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, and Openness to Experience). All responses were made on a 5-point Likert-type scale from 1 (strongly disagree) to 5 (strongly agree) and each HEXACO trait was assessed using 16 items, with an additional 4 items used to assess the interstitial facet of Altruism.
Sampling strategy	We use a dataset collected in a previous study. Sampling is described in the original study.
Data collection	Data was collected in a previous study and its collection is described in Lee and Ashton (2018): Lee, K., & Ashton, M. C. (2018). Psychometric Properties of the HEXACO-100. Assessment, 25(5), 1-15. https://doi.org/10.1177/1073191116659134
Timing	We use a dataset collected in a previous study. Sampling is described in the original study.
Data exclusions	We used only the data from the 100,318 (of 104,467) who completed all 100 items from the HEXACO-100 and correctly responded to three attentiveness-check items. Details regarding data collection and data cleansing are reported in Lee and Ashton (2018). The additional independent dataset (n = 270,539) was provided to us after data cleansing.
Non-participation	N/A
Randomization	N/A

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Ethics oversight	Detailed in Lee and Ashton (2018). Provided by the research ethics board at the University of Calgary.

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